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Completed

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Exc. No: 01

Networking commands and Uses in Linux and Windows

Aim:
Study of various Network commands used in Linux and Windows.

Basic Networking commands - Windows :-

1. **arp -a :** (device discovery)
→ address resolution protocol, shows IP address of computer and connected devices and MAC address of router.

2. **hostname:**
→ name of our own computer (host)

3. **ipconfig /all:**
→ detailed configuration information (networ setup): routers, gateways, DNS, DHCP and type of Ethernet

4. **netstat -a :** (device discovery)
→ helps to solve problems with NetBIOS name resolution (troubleshoot name issues, older window network naming)

5. **nslookup:**
→ name server lookup - DNS lookups and display DNS details like IP address both in interactive and non-interactive.

6. **pathping:**
→ unique to windows, shows the path our data takes to reach the destination and find data loss for slow connection identification.
Eg: ping localhost / 4.2.2.2 / ibaan.

connectivity.

10. tracert (Windows) and traceroute (Linux)
→ shows path (hop) our data takes to reach destination.

11. telnet: tests if a specific portion of a remote device is open (reachable).

IP & ~~Address~~ Adapter Config:-

12. ipconfig: shows/manages network adapter IP setting.

13. getmac: shows MAC address of network adapter.

DNS tools:

lookup:

* dig: flexible version of nslookup
* host.

14. iperf / iperf 3: measures actual bandwidth/ throughput between 2 devices.

15. netsh: advanced tool to view/config network settings.

Networking Commands - Linux

1. **ip**:
   → View and manages network settings (show IP, add & remove IP, enable and disable network interfaces and manage router).

a) Show IP address:
[root @ server ~] # ip address show.

b) add an IP address :- assign new IP address
ip address add 192.168.1.254/24

c) delete an IP address:
ip address del 192.168.1.254/24 dev enp0s3.

d) enables a network interface - turn on network connection: # ip link set eth0 up.

e) disable a network:
ip link set eth0 down.

f) enable promiscuous mode: allow interfaces recieve all net packets # ip link set eth0 promisc on

g) Add a route: adds path to reach another network
ip route add 192.168.1.0/24 via 192.168.1.254.

h) delete a route:
ip route delete 192.168.1.0/24 via 192.168.1.254

2. **ip config**:
   → config and display network interfaces.
(IP & status of network interfaces).

3.

3. mtr:
→ checks path between computers (ping and traceroute combination). It shows packet loss delay and route along the path.
a) mtr google.com.
→ every router between computer and google.
b) mtr -g google.com.
→ router IP address instead of names
c) mtr -c 10 google.com.
→ sends 10 test packets.

4. tcpdump :-
→ captures and displays network packets

Install
#dnf install -y tcpdump
show available interfaces
tcpdump -D
capture traffic : all packets on interface eth
#tcpdump -i eth0

Outgoing traffic:
#tcpdump -i eth0 dst 8.8.8.8

DNS traffic:
#tcpdump -i eth0 port 53.

HTTPS traffic:
#tcpdump -i eth0 www.google.com and port 443.

5. Ping

→ test whether another computer or website is reachable

ping google.com

ping 10 times

ping -c 10 google.com

common messages:

* Destination host unreachable: no route exist
* Request timed out: no reply received.
* Unknown host: Incorrect host or doesn't exist

6. Configure IPV6:

Automatic

nmcli connection modify "Wired connection 1"

*ipv6 meathod auto.

Static

manual.

7. Activate connection:

nmcli connection up Internal-LAN

IP address: ip address show

Default gateway: "route"

IPV6 gateway: ip -6 route show

DNC servers: cat /etc/resolv.conf

Test network: ping google.com

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